

Lista 3

① $A = \begin{pmatrix} 16 & 4 & 8 & 4 \\ 4 & 10 & 8 & 4 \\ 8 & 8 & 12 & 10 \\ 4 & 4 & 10 & 12 \end{pmatrix}$, $b = \begin{pmatrix} 32 \\ 26 \\ 28 \\ 30 \end{pmatrix}$

② $\text{Det}(A) = 16 \cdot \begin{vmatrix} 10 & 8 & 4 \\ 8 & 12 & 10 \\ 4 & 10 & 12 \end{vmatrix} - (4) \cdot \begin{vmatrix} 4 & 8 & 4 \\ 8 & 12 & 10 \\ 4 & 10 & 12 \end{vmatrix} + 8 \cdot \begin{vmatrix} 4 & 10 & 4 \\ 8 & 8 & 10 \\ 4 & 4 & 12 \end{vmatrix} -$

$4 \cdot \begin{vmatrix} 4 & 10 & 8 \\ 8 & 8 & 12 \\ 4 & 4 & 10 \end{vmatrix}$

$16 \cdot \left(10 \cdot \begin{vmatrix} 12 & 10 \\ 10 & 12 \end{vmatrix} - 8 \cdot \begin{vmatrix} 8 & 10 \\ 4 & 12 \end{vmatrix} + 4 \cdot \begin{vmatrix} 8 & 12 \\ 4 & 10 \end{vmatrix} \right) = 1920$
 $-100 + 144 \quad -40 + 96 \quad -48 + 80$
 $440 - 448 + 128$

$-4 \cdot \left(4 \cdot \begin{vmatrix} 12 & 10 \\ 10 & 12 \end{vmatrix} - 8 \cdot \begin{vmatrix} 8 & 10 \\ 4 & 12 \end{vmatrix} + 4 \cdot \begin{vmatrix} 8 & 12 \\ 4 & 10 \end{vmatrix} \right) = 576$
 $-100 + 144 \quad -40 + 96 \quad -48 + 80$
 $440 - 448 + 128$

$8 \cdot \left(4 \cdot \begin{vmatrix} 8 & 10 \\ 4 & 12 \end{vmatrix} - 10 \cdot \begin{vmatrix} 8 & 10 \\ 4 & 12 \end{vmatrix} + 4 \cdot \begin{vmatrix} 8 & 8 \\ 4 & 4 \end{vmatrix} \right) = -2,688$
 $224 \quad 56 \quad 0$
 -560

$-4 \cdot \left(4 \cdot \begin{vmatrix} 8 & 12 \\ 4 & 10 \end{vmatrix} - 10 \cdot \begin{vmatrix} 8 & 12 \\ 4 & 10 \end{vmatrix} + 0 \right) = 768$
 $128 \quad 3680 \quad -48 + 80 = -320 = -320 + 1201 \cdot 4$

$\text{Det}(A) = 576$

$$\textcircled{b} \det(A_{11}) = 16 \neq 0$$

$$\det \begin{pmatrix} 16 & 4 \\ 4 & 10 \end{pmatrix} = 160 - 16 = 144$$

$$\det \begin{pmatrix} 16 & 4 & 8 \\ 4 & 10 & 8 \\ 8 & 8 & 12 \end{pmatrix} = 576$$

$$\det(A) = 576$$

ii) Calc U_{11}

$$L = \begin{pmatrix} 1 & 0 & 0 & 0 \\ \frac{1}{4} & 1 & 0 & 0 \\ \frac{1}{2} & \frac{2}{3} & 1 & 0 \\ \frac{1}{4} & \frac{1}{3} & \frac{3}{2} & 1 \end{pmatrix}$$

$$U = \begin{pmatrix} 16 & 4 & 8 & 4 \\ 0 & 9 & 6 & 3 \\ 0 & 0 & 4 & 6 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$$L \cdot y = b$$

$$\therefore \begin{pmatrix} 1 & 0 & 0 & 0 \\ \frac{1}{4} & 1 & 0 & 0 \\ \frac{1}{2} & \frac{2}{3} & 1 & 0 \\ \frac{1}{4} & \frac{1}{3} & \frac{3}{2} & 1 \end{pmatrix} \cdot \begin{pmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \end{pmatrix} = \begin{pmatrix} 32 \\ 26 \\ 38 \\ 30 \end{pmatrix}$$

$$y_1 = 32;$$

$$\frac{1}{4} \cdot 32 + y_2 = 26 \therefore y_2 = 18$$

$$\frac{1}{2} \cdot 32 + \frac{2}{3} \cdot 18 + y_3 = 38 \Rightarrow 16 + \frac{36}{3} + y_3 = 38 \therefore y_3 = 10$$

$$y_4 = 1$$

$$b) \quad y = \begin{pmatrix} 32 \\ 18 \\ 10 \\ 1 \end{pmatrix}$$

$$Ux = y \quad U = \begin{pmatrix} 16 & 4 & 8 & 4 \\ 0 & 9 & 6 & 3 \\ 0 & 0 & 4 & 6 \\ 0 & 0 & 0 & 1 \end{pmatrix} \cdot \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} = \begin{pmatrix} 32 \\ 18 \\ 10 \\ 1 \end{pmatrix}$$

$$x_4 = 1; \\ 4x_3 + 6 = 10 \therefore \underline{x_3 = 1}; \quad 9x_2 + 6 + 3 = 18 \therefore x_2 = 1$$

$$x_1 = 1 \\ x = \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix}$$

$$c) \quad \det(U) = 1 \cdot \begin{vmatrix} 16 & 4 & 8 \\ 0 & 9 & 6 \\ 0 & 0 & 4 \end{vmatrix} = 1 \cdot (4 \cdot (16 \cdot 9)) = 576$$

$$\det(U) = \det(A) \therefore \underline{\det(A) = 576} //$$

$$2) \quad A = \begin{pmatrix} 2 & 1 & 3 \\ 0 & -1 & 1 \\ 1 & 0 & 3 \end{pmatrix}, \quad b = \begin{pmatrix} 9 \\ 1 \\ 7 \end{pmatrix}$$

$$a) \quad \det(A_1) = 2 \neq 0$$

$$\det(A_2) = \begin{vmatrix} 2 & 1 \\ 0 & -1 \end{vmatrix} \neq 0$$

$$\det(A_3) = 1 \cdot \begin{vmatrix} 1 & 3 \\ -1 & 1 \end{vmatrix} - 0 + 3 \cdot \begin{vmatrix} 2 & 1 \\ 0 & -1 \end{vmatrix} \neq 0$$

Calculando L e U

$$A = \begin{pmatrix} 2 & 1 & 3 \\ 0 & -1 & 1 \\ 1 & 0 & 3 \end{pmatrix} \xrightarrow{A^{(0)}} \begin{pmatrix} 2 & 1 & 3 \\ 0 & -1 & 1 \\ 0 & -\frac{1}{2} & \frac{3}{2} \end{pmatrix} \xrightarrow{} \begin{pmatrix} 2 & 1 & 3 \\ 0 & -1 & 1 \\ 0 & 0 & 1 \end{pmatrix}$$

$$L = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ \frac{1}{2} & \frac{1}{2} & 1 \end{pmatrix}$$

$$3 - \frac{3}{2} = \frac{6-3}{2} = \frac{3}{2}$$
$$\frac{3}{2} - \frac{1}{2} = \frac{2}{2} = 1$$

$$U = \begin{pmatrix} 2 & 1 & 3 \\ 0 & -1 & 1 \\ 0 & 0 & 1 \end{pmatrix}$$

$$L \cdot y = b$$

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ \frac{1}{2} & \frac{1}{2} & 1 \end{pmatrix} \cdot \begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix} = \begin{pmatrix} 9 \\ 1 \\ 7 \end{pmatrix}$$

$$y_1 = 9$$

$$y_2 = 1$$

$$y_3 = -\left(\frac{9}{2} + \frac{1}{2}\right) + 7 = 2$$

$$y = (9, 1, 2)^T$$

$$U \cdot x = y$$

$$\begin{pmatrix} 2 & 1 & 3 \\ 0 & -1 & 1 \\ 0 & 0 & 1 \end{pmatrix} \cdot \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 9 \\ 1 \\ 2 \end{pmatrix}$$

$$x_3 = 2$$

$$-x_2 + 2 = 1 \quad x_2 = 1$$

$$2x_3 + 1 + 6 = 9 \Rightarrow \frac{2}{2} = 1$$

$$x = \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix}$$

③

$$\begin{cases} u_{ij} = a_{ij} - \sum_{k=1}^{i-1} l_{ik} u_{kj} & i \leq j \\ l_{ij} = \frac{a_{ij} - \sum_{k=1}^{j-1} l_{ik} u_{kj}}{u_{jj}} & i > j \end{cases}$$

④ $Ax = b$

$$A = \begin{pmatrix} 2 & 3 & -1 \\ 1 & 0 & 2 \\ 0 & 3 & -1 \end{pmatrix}, \quad b = \begin{pmatrix} 4 \\ 3 \\ 2 \end{pmatrix}$$

i)

$$\text{Det}(A_1) = 2; \text{Det}(A_2) = -3; \text{Det}(A_3) = 2 \cdot \begin{vmatrix} 0 & 2 \\ 3 & -1 \end{vmatrix} - 1 \cdot \begin{vmatrix} 3 & -1 \\ 3 & -1 \end{vmatrix} + 0$$

$$\text{Det}(A_3) = -12 \neq 0$$

ii)

$$A^{(1)} = \begin{pmatrix} 2 & 3 & -1 \\ 1 & 0 & 2 \\ 0 & 3 & -1 \end{pmatrix} \xrightarrow{(x \frac{1}{2})} \begin{pmatrix} 2 & 3 & -1 \\ 0 & -\frac{3}{2} & \frac{5}{2} \\ 0 & 3 & -1 \end{pmatrix} \xrightarrow{(x-2)} \begin{pmatrix} 2 & 3 & -1 \\ 0 & -\frac{3}{2} & \frac{5}{2} \\ 0 & 0 & 4 \end{pmatrix} A^{(2)}$$

$$L = \begin{pmatrix} 1 & 0 & 0 \\ \frac{1}{2} & 1 & 0 \\ 0 & -2 & 1 \end{pmatrix}$$

$$U = \begin{pmatrix} 2 & 3 & -1 \\ 0 & -\frac{3}{2} & \frac{5}{2} \\ 0 & 0 & 4 \end{pmatrix}$$

iii) $Ly = b$

$$\begin{pmatrix} 1 & 0 & 0 \\ \frac{1}{2} & 1 & 0 \\ 0 & -2 & 1 \end{pmatrix} \cdot \begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix} = \begin{pmatrix} 4 \\ 3 \\ 2 \end{pmatrix} \quad \begin{cases} y_1 = 4 \\ y_2 = +1 \\ y_3 = 4 \end{cases}$$

$$y = \begin{pmatrix} 4 \\ 1 \\ 4 \end{pmatrix}$$

$$4 \geq 4 y_3 = 2$$

2

iii) $Ux = y$

$$\begin{pmatrix} 2 & 3 & -1 \\ 0 & -\frac{3}{2} & \frac{5}{2} \\ 0 & 0 & 4 \end{pmatrix} \cdot \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 4 \\ 1 \\ 4 \end{pmatrix}$$

$x_3 = 1$, $x_3 = 1$ $\therefore -\frac{3}{2}x_2 + \frac{5}{2} = 1 \Rightarrow -\frac{3}{2}x_2 = 1 - \frac{5}{2} = -\frac{3}{2} \Rightarrow x_2 = 1$

$2x_1 + 1 = 4 \Rightarrow 2x_1 = 3 \Rightarrow x_1 = \frac{3}{2}$

$x = \begin{pmatrix} \frac{3}{2} \\ 1 \\ 1 \end{pmatrix}$

$-3x_2 + 5 = 2 \Rightarrow -3x_2 = 2 - 5 = -3 \Rightarrow x_2 = 1$

(5)

(5.1)

$$10 - \frac{3}{2} = \frac{17}{2} - \frac{2}{3} = \frac{51 - 4}{6} = \frac{47}{6}$$

$$(A_1 | b_1) = \left(\begin{array}{ccc|c} 2 & -1 & 3 & -4 \\ 4 & 1 & 2 & -7 \\ 1 & 0 & 1 & -1 \end{array} \right) = \left(\begin{array}{ccc|c} 2 & -1 & 3 & -4 \\ 0 & 3 & -4 & 1 \\ 0 & 1 & \frac{17}{2} & -9 \end{array} \right) = \left(\begin{array}{ccc|c} 2 & -1 & 3 & -4 \\ 0 & 3 & -4 & 1 \\ 0 & 0 & \frac{55}{6} & -\frac{55}{6} \end{array} \right)$$

$x_3 = -1$ $\therefore 3x_2 + 4 = 1 \Rightarrow x_2 = -1$ $x_1 = -1$

$x = (-1, -1, -1)$

$2x_1 + 2 - 3 = -4$

(5.2)

$$(A_2 | b_2) = \left(\begin{array}{ccc|c} 2 & -1 & 3 & 2 \\ 0 & 3 & -4 & 2 \\ 0 & 0 & \frac{55}{6} & \frac{2}{3} \end{array} \right)$$

$y_1 = \frac{14}{11}$, $y_2 = \frac{42}{55}$, $y_3 = \frac{4}{55}$

$y = \begin{pmatrix} \frac{14}{11} \\ \frac{42}{55} \\ \frac{4}{55} \end{pmatrix}$



$$2 \cdot \frac{1}{6} = \frac{1}{3}$$

$$(A_3 | b_3) = \left(\begin{array}{ccc|c} 2 & -1 & 3 & 4 \\ 0 & 3 & -4 & -2 \\ 0 & 0 & \frac{55}{6} & \frac{155}{3} \end{array} \right) \begin{array}{l} -8+6=-2 \\ -2+20=18 \end{array}$$

$$z_3 = \frac{6 \cdot \frac{155}{3}}{3 \cdot \frac{55}{6}} = \frac{2}{1} \quad z_2 = 2 \quad z_1 = 0$$

$$z = \begin{pmatrix} 0 \\ 2 \\ 2 \end{pmatrix}$$

5b)

$$(A|b) = \left(\begin{array}{ccc|c} 10 & 1 & -1 & 10 \\ 1 & 10 & 1 & 12 \\ 2 & -1 & 10 & 11 \end{array} \right) = \left(\begin{array}{ccc|c} 10 & 1 & -1 & 10 \\ 0 & \frac{99}{10} & \frac{11}{10} & 11 \\ 0 & -\frac{6}{5} & \frac{51}{5} & 9 \end{array} \right)$$

$$m_{11} = \frac{1}{10}, m_2 = \frac{1}{5}, m_3 = -\frac{4}{3}$$

$$x_3 = +1$$

$$= \left(\begin{array}{ccc|c} 10 & 1 & -1 & 10 \\ 0 & \frac{99}{10} & \frac{11}{10} & 11 \\ 0 & 0 & \frac{31}{3} & \frac{131}{3} \end{array} \right)$$

$$\frac{99}{10} x_2 + \frac{11}{10} = 11 \Rightarrow \frac{99}{10} x_2 = \frac{99}{10} \Rightarrow x_2 = 1$$

$$x_1 = 1$$

$$x = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$$

⑥

$$A = \begin{pmatrix} 8 & 1 & 2 \\ 1 & 1 & 1 \\ 2 & 1 & 3 \end{pmatrix}, \quad b = \begin{pmatrix} -1 \\ 1 \\ 4 \end{pmatrix}$$

⑦

$$A = LL^T \quad L = \begin{pmatrix} g_{11} & 0 & 0 \\ g_{21} & g_{22} & 0 \\ g_{31} & g_{32} & g_{33} \end{pmatrix}, \quad L^T = \begin{pmatrix} g_{11} & 0 & 0 \\ 0 & g_{22} & 0 \\ 0 & 0 & g_{33} \end{pmatrix}$$

$$g_{11} = \sqrt{5}; \quad g_{21} = \frac{1}{\sqrt{5}} = \frac{\sqrt{5}}{5}; \quad g_{31} = \frac{2\sqrt{5}}{5}$$

$$L = \begin{pmatrix} \sqrt{5} & 0 & 0 \\ \frac{\sqrt{5}}{5} & \frac{2\sqrt{5}}{5} & 0 \\ \frac{2\sqrt{5}}{5} & \frac{3\sqrt{5}}{10} & \sqrt{\frac{7}{2}} \end{pmatrix}$$

$$a_{22} = g_{21}^2 + g_{22}^2$$

$$g_{22} = \sqrt{a_{22} - g_{21}^2}$$

$$a_{32} = g_{21}g_{31} + g_{22}g_{32}$$

$$g_{32} = \sqrt{1 - \frac{1}{5}}$$

i) $\det(L) = \begin{vmatrix} \frac{2\sqrt{5}}{5} & 0 \\ \frac{2}{\sqrt{5}} & \frac{\sqrt{7}}{2} \end{vmatrix} \cdot \sqrt{5} = \frac{5}{2\sqrt{5}} \cdot \frac{5\sqrt{5}}{10} = \frac{5}{2\sqrt{5}} \cdot \frac{\sqrt{5}}{2} = \frac{5}{4}$

$$\det(L) = \frac{2\sqrt{7}}{5} \cdot \sqrt{5} = \sqrt{7}$$

$$\therefore \det(A) = \det(L)^2 = 7$$

$$Ly = b \quad \therefore \begin{pmatrix} \sqrt{5} & 0 & 0 \\ \frac{\sqrt{5}}{5} & \frac{2\sqrt{5}}{5} & 0 \\ \frac{2\sqrt{5}}{5} & \frac{3\sqrt{5}}{10} & \sqrt{\frac{7}{2}} \end{pmatrix} \cdot \begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix} = \begin{pmatrix} -1 \\ 1 \\ 4 \end{pmatrix}$$

$$y_1 \sqrt{5} = -1 \quad \therefore y_1 = \frac{-1}{\sqrt{5}}$$

$$-\frac{\sqrt{5}}{5} \cdot \frac{1}{\sqrt{5}} + \frac{2\sqrt{5}}{5} y_2 = 1 \quad \therefore \frac{-1}{\sqrt{5}} + \frac{2\sqrt{5}}{5} y_2 = 1$$

$$y_2 = \frac{3}{\sqrt{5}} = \frac{3\sqrt{5}}{5}$$

$$-\frac{2}{\sqrt{5}} + \frac{9}{10} + \frac{\sqrt{7}}{2} y_3 = 4 \quad y_3 = \frac{7}{2\sqrt{2}}$$

$$y = \begin{pmatrix} -\frac{1}{\sqrt{5}} \\ \frac{3}{\sqrt{5}} \\ \sqrt{7} \end{pmatrix} \Rightarrow L^+ \cdot x = y$$

$$\begin{pmatrix} \sqrt{5} & \frac{\sqrt{5}}{5} & \frac{2\sqrt{5}}{5} \\ 0 & \frac{2\sqrt{5}}{5} & \frac{3\sqrt{5}}{10} \\ 0 & 0 & \frac{\sqrt{7}}{2} \end{pmatrix} \cdot \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} -\frac{1}{\sqrt{5}} \\ \frac{3}{\sqrt{5}} \\ \sqrt{7} \end{pmatrix}$$

$$x_3 \cdot \frac{\sqrt{7}}{2} = \sqrt{7} \Rightarrow x_3 = 2 \quad \left| \begin{array}{l} \frac{2\sqrt{5}}{5} \cdot x_2 + \frac{6\sqrt{5}}{10} = \frac{3}{\sqrt{5}} \end{array} \right.$$

$$\frac{2\sqrt{5}}{5} x_2 = \frac{3}{\sqrt{5}} - \frac{6\sqrt{5}}{10} \Rightarrow \frac{30 - 30}{10\sqrt{5}} = 0 \Rightarrow x_2 = 0$$

$$x_1 \cdot \sqrt{5} + \frac{\sqrt{5}}{5} \cdot 0 + \frac{2\sqrt{5}}{5} \cdot 2 = -\frac{1}{\sqrt{5}} \quad \left| \begin{array}{l} x_1 = -\frac{1}{5} \end{array} \right.$$

$$\vec{x} = \begin{pmatrix} -\frac{1}{5} \\ 0 \\ 2 \end{pmatrix}$$

a)

$$\begin{pmatrix} 1 & 2 & 4 & 1 & | & 1 \\ 2 & 1 & -1 & 3 & | & -6 \\ -1 & 3 & 1 & -5 & | & 7 \\ 3 & 4 & 2 & 6 & | & 13 \end{pmatrix} = \begin{pmatrix} 1 & 2 & 4 & 1 & | & 1 \\ 0 & -3 & -9 & 1 & | & -8 \\ 0 & 5 & 5 & -4 & | & 8 \\ 0 & -2 & -10 & -9 & | & 10 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} 1 & 2 & 4 & 1 & | & 1 \\ 0 & -1 & 3 & -\frac{1}{3} & | & -\frac{8}{3} \\ 0 & 5 & 5 & -4 & | & 8 \\ 0 & -2 & -10 & -9 & | & 10 \end{pmatrix} \Rightarrow \begin{pmatrix} 1 & 0 & -2 & \frac{5}{3} & | & -\frac{13}{3} \\ 0 & 1 & 3 & -\frac{1}{3} & | & \frac{8}{3} \\ 0 & 0 & -10 & -\frac{7}{3} & | & -\frac{16}{3} \\ 0 & 0 & -4 & -\frac{29}{3} & | & \frac{46}{3} \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} 1 & 0 & -2 & \frac{5}{3} & | & -\frac{13}{3} \\ 0 & 1 & 3 & -\frac{1}{3} & | & \frac{8}{3} \\ 0 & 0 & -1 & -\frac{7}{30} & | & \frac{8}{15} \\ 0 & 0 & -4 & -\frac{29}{3} & | & \frac{46}{3} \end{pmatrix} \Rightarrow \begin{pmatrix} 1 & 0 & 0 & \frac{32}{15} & | & -\frac{49}{15} \\ 0 & 1 & 0 & -\frac{31}{30} & | & \frac{16}{15} \\ 0 & 0 & 1 & \frac{7}{30} & | & \frac{8}{15} \\ 0 & 0 & 0 & -\frac{131}{15} & | & \frac{262}{15} \end{pmatrix}$$

$$\begin{pmatrix} 1 & 0 & 0 & \frac{32}{15} & | & -\frac{49}{15} \\ 0 & 1 & 0 & -\frac{31}{30} & | & \frac{16}{15} \\ 0 & 0 & 1 & \frac{7}{30} & | & \frac{8}{15} \\ 0 & 0 & 0 & 1 & | & -2 \end{pmatrix} \Rightarrow \begin{pmatrix} 1 & 0 & 0 & 0 & | & 1 \\ 0 & 1 & 0 & 0 & | & -1 \\ 0 & 0 & 1 & 0 & | & 1 \\ 0 & 0 & 0 & 1 & | & -2 \end{pmatrix}$$

$$x = \begin{pmatrix} 1 \\ -1 \\ 1 \\ -2 \end{pmatrix}$$

||

$$(A|b) = \left(\begin{array}{cccc|c} 2 & 4 & 6 & 1 & -8 \\ 1 & -3 & -1 & 2 & 4 \\ 2 & 1 & 3 & -1 & -2 \\ 4 & -2 & 1 & -4 & 6 \end{array} \right) \xrightarrow{L_2 \leftrightarrow L_1} \left(\begin{array}{cccc|c} 1 & -3 & -1 & 2 & 4 \\ 2 & 4 & 6 & 1 & -8 \\ 2 & 1 & 3 & -1 & -2 \\ 4 & -2 & 1 & -4 & 6 \end{array} \right)$$

$$= \left(\begin{array}{cccc|c} 1 & -3 & -1 & 2 & 4 \\ 0 & 10 & 8 & -3 & -16 \\ 0 & 7 & 5 & -5 & -10 \\ 0 & 10 & 5 & -12 & -10 \end{array} \right) \xrightarrow{\frac{6}{5}} \left(\begin{array}{cccc|c} 1 & -3 & -1 & 2 & 4 \\ 0 & 1 & \frac{1}{2} & -\frac{6}{5} & -\frac{1}{5} \\ 0 & 7 & 5 & -5 & -10 \\ 0 & 10 & 8 & -3 & -16 \end{array} \right)$$

$$= \left(\begin{array}{cccc|c} 1 & -3 & -1 & 2 & 4 \\ 0 & 1 & \frac{1}{2} & -\frac{6}{5} & -\frac{1}{5} \\ 0 & 0 & \frac{3}{2} & \frac{16}{5} & -3 \\ 0 & 0 & 3 & 5 & -6 \end{array} \right) \Rightarrow \left(\begin{array}{cccc|c} 1 & 0 & \frac{1}{2} & -\frac{8}{5} & 1 \\ 0 & 1 & \frac{1}{2} & -\frac{6}{5} & -1 \\ 0 & 0 & \frac{1}{2} & 3 & -2 \\ 0 & 0 & \frac{3}{2} & 16 & -3 \end{array} \right)$$

$$= \left(\begin{array}{cccc|c} 1 & 0 & 0 & -\frac{31}{10} & 2 \\ 0 & 1 & 0 & -\frac{27}{10} & 0 \\ 0 & 0 & 1 & 3 & -2 \\ 0 & 0 & 0 & \frac{23}{2} & 0 \end{array} \right) \xrightarrow{\begin{matrix} -2 \cdot \frac{2}{3} \\ +\frac{4}{3} - 3 \end{matrix}} \left(\begin{array}{cccc|c} 1 & 0 & 0 & 0 & \frac{26}{3} \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & -2 \\ 0 & 0 & 0 & 1 & 0 \end{array} \right)$$

$$x = \begin{pmatrix} 2 \\ 0 \\ -2 \\ 0 \end{pmatrix}$$

⑧

$$A = \begin{pmatrix} 2 & 1 & -1 \\ 1 & 10 & 2 \\ -1 & 2 & 4 \end{pmatrix} \quad | \quad L L^T$$

$$= \begin{pmatrix} l_{11}^2 & l_{21}l_{11} & l_{31}l_{11} \\ l_{11}l_{21} & l_{21}^2 + l_{22}^2 & l_{21}l_{31} + l_{22}l_{32} \\ l_{11}l_{31} & l_{31}l_{21} + l_{32}l_{22} & l_{31}^2 + l_{32}^2 + l_{33}^2 \end{pmatrix}$$

1st column

$$l_{11} = \sqrt{21}$$

$$l_{21} = \frac{\sqrt{21}}{2}$$

$$l_{31} = -\frac{\sqrt{21}}{2}$$

$$l_{22} = \sqrt{\frac{19}{2}}$$

$$l_{32} = \frac{5\sqrt{38}}{38}$$

$$l_{33} = \sqrt{\frac{1020}{301}}$$

$$L y = I$$

$$\begin{pmatrix} \sqrt{21} & 0 & 0 \\ \frac{\sqrt{21}}{2} & \sqrt{\frac{19}{2}} & 0 \\ -\frac{\sqrt{21}}{2} & \frac{5\sqrt{38}}{38} & \sqrt{\frac{1020}{301}} \end{pmatrix} \cdot \begin{pmatrix} y_{11} & y_{12} & y_{13} \\ y_{21} & y_{22} & y_{23} \\ y_{31} & y_{32} & y_{33} \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$y = \begin{pmatrix} \frac{\sqrt{21}}{2} & -\frac{\sqrt{38}}{38} & \frac{2\sqrt{114}}{57} \\ 0 & \frac{\sqrt{38}}{19} & -\frac{5\sqrt{114}}{342} \\ 0 & 0 & \frac{\sqrt{114}}{18} \end{pmatrix}$$

$$L^T x = y$$

$$\begin{pmatrix} \sqrt{21} & \frac{\sqrt{21}}{2} & -\frac{\sqrt{21}}{2} \\ 0 & \sqrt{\frac{19}{2}} & \frac{5\sqrt{38}}{38} \\ 0 & 0 & \frac{5\sqrt{114}}{19} \end{pmatrix} \cdot \begin{pmatrix} x_{11} & x_{12} & x_{13} \\ x_{21} & x_{22} & x_{23} \\ x_{31} & x_{32} & x_{33} \end{pmatrix} = \begin{pmatrix} \frac{\sqrt{21}}{2} & \frac{\sqrt{38}}{38} & \frac{2\sqrt{114}}{57} \\ 0 & \frac{\sqrt{38}}{19} & -\frac{5\sqrt{114}}{342} \\ 0 & 0 & \frac{\sqrt{114}}{18} \end{pmatrix}$$

$$x = \begin{pmatrix} \frac{2}{3} & -\frac{1}{9} & \frac{2}{9} \\ -\frac{1}{9} & \frac{7}{54} & -\frac{5}{54} \\ \frac{2}{9} & -\frac{5}{34} & \frac{19}{54} \end{pmatrix} //$$

y_2

$$L \cdot \begin{pmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix} \Rightarrow \begin{pmatrix} 0 \\ 1 \\ -7/11 \\ -11/17 \end{pmatrix}$$

y_3

$$L \cdot \begin{pmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix} \Rightarrow \begin{pmatrix} 0 \\ 0 \\ 1 \\ 4/17 \end{pmatrix}$$

y_4

$$L \cdot \begin{pmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}$$

$$y = \begin{pmatrix} 1 & 0 & 0 & 0 \\ -1/11 & 1 & 0 & 0 \\ -7/11 & -7/11 & 1 & 0 \\ -5/17 & -11/17 & 4/17 & 1 \end{pmatrix}$$

Calculando $Ux = y$

$$\begin{pmatrix} 4 & 3 & 2 & 1 \\ 0 & -1/4 & 5/2 & 15/4 \\ 0 & 0 & -3/11 & -10/11 \\ 0 & 0 & 0 & 4/17 \end{pmatrix} \cdot \begin{pmatrix} x_{11} \\ x_{21} \\ x_{31} \\ x_{41} \end{pmatrix} = \begin{pmatrix} 1 \\ -1/4 \\ -1/11 \\ -5/17 \end{pmatrix} \Rightarrow \begin{pmatrix} 0 \\ -1/4 \\ 3/2 \\ -5/4 \end{pmatrix}$$

$$U \cdot \vec{x}_2 = \vec{y}_2 \Rightarrow \begin{pmatrix} 0 \\ -1/4 \\ 1/2 \\ -1/4 \end{pmatrix}$$

$$U \cdot \vec{x}_3 = \vec{y}_3 \Rightarrow \begin{pmatrix} 1/2 \\ 0 \\ -3/2 \\ 1 \end{pmatrix}$$

$$U \cdot \vec{x}_4 = \vec{y}_4 \Rightarrow \begin{pmatrix} 1/2 \\ 5/4 \\ -5 \\ 17/4 \end{pmatrix}$$

$$X = \begin{pmatrix} 0 & 0 & 1/2 & 1/2 \\ -1/4 & -1/4 & 0 & 5/4 \\ 3/2 & 1/2 & -3/2 & -5 \\ -5/4 & -1/4 & 1 & 17/4 \end{pmatrix}$$

a)

$$A = \begin{pmatrix} 4 & 2 & -4 \\ 2 & 10 & 4 \\ -4 & 4 & 9 \end{pmatrix} \Rightarrow \left(\begin{array}{ccc|ccc} 4 & 2 & -4 & 1 & 0 & 0 \\ 0 & 9 & 6 & -\frac{1}{2} & 1 & 0 \\ 0 & 6 & 5 & 1 & 0 & 1 \end{array} \right)$$

$$= \left(\begin{array}{ccc|ccc} 4 & 2 & -4 & 1 & 0 & 0 \\ 0 & 9 & 6 & -\frac{1}{2} & 1 & 0 \\ 0 & 0 & 1 & \frac{4}{3} & -\frac{2}{3} & 1 \end{array} \right) \Rightarrow \left(\begin{array}{ccc|ccc} 1 & \frac{1}{2} & -1 & \frac{1}{4} & 0 & 0 \\ 0 & 1 & \frac{2}{3} & -\frac{1}{18} & \frac{1}{9} & 0 \\ 0 & 0 & 1 & \frac{4}{3} & -\frac{2}{3} & 1 \end{array} \right)$$

$$\Rightarrow \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & \frac{13}{12} & -\frac{1}{2} & \frac{1}{4} \\ 0 & 1 & 0 & -\frac{3}{4} & \frac{1}{2} & -\frac{2}{3} \\ 0 & 0 & 1 & \frac{4}{3} & -\frac{2}{3} & 1 \end{array} \right) \therefore A^{-1} = \begin{pmatrix} \frac{13}{12} & -\frac{1}{2} & \frac{1}{4} \\ -\frac{3}{4} & \frac{1}{2} & -\frac{2}{3} \\ \frac{4}{3} & -\frac{2}{3} & 1 \end{pmatrix}$$

b)

$$A = \begin{pmatrix} 2 & 4 & 6 \\ 1 & -3 & -1 \\ 2 & 1 & 1 \end{pmatrix} \Rightarrow \left(\begin{array}{ccc|ccc} 1 & 2 & 3 & \frac{1}{2} & 0 & 0 \\ 0 & -5 & -4 & -\frac{1}{2} & 1 & 0 \\ 0 & -3 & -5 & -1 & 0 & 1 \end{array} \right)$$

$$= \left(\begin{array}{ccc|ccc} 1 & 2 & 3 & \frac{1}{2} & 0 & 0 \\ 0 & 1 & \frac{4}{5} & +\frac{1}{10} & -\frac{1}{5} & 0 \\ 0 & -3 & 5 & -1 & 0 & 1 \end{array} \right) \Rightarrow \left(\begin{array}{ccc|ccc} 1 & 0 & \frac{7}{5} & \frac{3}{10} & \frac{2}{5} & 0 \\ 0 & 1 & \frac{4}{5} & \frac{1}{10} & -\frac{1}{5} & 0 \\ 0 & 0 & 1 & -\frac{7}{10} & -\frac{3}{10} & \frac{5}{10} \end{array} \right)$$

$$= \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & -\frac{1}{13} & \frac{4}{13} & \frac{7}{13} \\ 0 & 1 & 0 & -\frac{3}{26} & -\frac{5}{13} & \frac{4}{13} \\ 0 & 0 & 1 & \frac{7}{13} & \frac{3}{13} & -\frac{5}{13} \end{array} \right) \therefore A^{-1} = \begin{pmatrix} -\frac{1}{13} & \frac{4}{13} & \frac{7}{13} \\ -\frac{3}{26} & -\frac{5}{13} & \frac{4}{13} \\ \frac{7}{13} & \frac{3}{13} & -\frac{5}{13} \end{pmatrix}$$

10a

$$A = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 1 & -2 & 3 & 4 \\ 1 & -1 & -1 & -1 \\ 1 & 1 & 1 & 1 \end{pmatrix}$$

Convergenz

$$\det(A_1) \neq 0$$

$$\det(A_2) = \begin{vmatrix} 4 & 3 \\ 1 & -2 \end{vmatrix} \neq 0 = -11$$

$$\det(A_3) = \begin{vmatrix} 4 & 3 & 2 \\ 1 & -2 & 3 \\ 1 & -1 & -1 \end{vmatrix} = 34 \neq 0$$

$$\det(A_4) \neq 0$$

Converge

$$L = \begin{pmatrix} 1 & 0 & 0 & 0 \\ L_{21} & 1 & 0 & 0 \\ L_{31} & L_{32} & 1 & 0 \\ L_{41} & L_{42} & L_{43} & 1 \end{pmatrix}$$

e

$$U = \begin{pmatrix} u_{11} & u_{12} & u_{13} & u_{14} \\ 0 & & & u_{24} \\ 0 & u_{22} & u_{23} & u_{34} \\ 0 & 0 & u_{33} & u_{44} \end{pmatrix}$$

$$u_{11} = 4 \quad L_{21} = \frac{1}{4} \quad u_{22} = -\frac{11}{4} \quad u_{23} = \frac{5}{2} \quad u_{24} = \frac{5}{4}$$

$$u_{12} = 3 \quad L_{31} = \frac{1}{4} \quad L_{32} = \frac{7}{11} \quad u_{33} = -\frac{34}{11} \quad u_{34} = -\frac{40}{11}$$

$$u_{13} = 2 \quad L_{41} = \frac{1}{4} \quad L_{42} = -\frac{1}{11} \quad L_{43} = -\frac{4}{17} \quad u_{44} = \frac{4}{17}$$

$$u_{14} = 1$$

$$L = \begin{pmatrix} 1 & 0 & 0 & 0 \\ \frac{1}{4} & 1 & 0 & 0 \\ \frac{1}{4} & \frac{7}{11} & 1 & 0 \\ \frac{1}{4} & -\frac{1}{11} & -\frac{4}{17} & 1 \end{pmatrix}$$

$$U = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 0 & -\frac{11}{4} & \frac{5}{2} & \frac{5}{4} \\ 0 & 0 & -\frac{34}{11} & -\frac{40}{11} \\ 0 & 0 & 0 & \frac{4}{17} \end{pmatrix}$$

$$L \vec{y} = \vec{b}$$

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ \frac{1}{4} & 1 & 0 & 0 \\ \frac{1}{4} & \frac{7}{11} & 1 & 0 \\ \frac{1}{4} & -\frac{1}{11} & -\frac{4}{17} & 1 \end{pmatrix} \cdot \begin{pmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} \Rightarrow \begin{pmatrix} 1 \\ -\frac{1}{4} \\ -\frac{1}{11} \\ -\frac{5}{12} \end{pmatrix}$$

$$A = \begin{pmatrix} 2 & 1 & -1 \\ 1 & 10 & 2 \\ -1 & 2 & 4 \end{pmatrix} \Rightarrow A^{(1)} = \begin{pmatrix} 2 & 1 & -1 \\ 0 & \frac{19}{2} & \frac{5}{2} \\ 0 & \frac{5}{2} & \frac{7}{2} \end{pmatrix} = A^{(1)} \begin{pmatrix} 2 & 1 & -1 \\ 0 & \frac{19}{2} & \frac{5}{2} \\ 0 & 0 & \frac{54}{19} \end{pmatrix}$$

$$m_{11} = \frac{1}{2}, m_{12} = -\frac{1}{2}$$

$$m_{22} = \frac{\frac{5}{2}}{\frac{19}{2}} = \frac{5}{19}$$

$$\frac{1}{2} + 4 = \frac{9}{2}$$

$$-\frac{1}{2} - 1 \quad 4 - \frac{1}{2} = \frac{7}{2}$$

$$\therefore L = \begin{pmatrix} 1 & 0 & 0 \\ \frac{1}{2} & 1 & 0 \\ -\frac{1}{2} & \frac{5}{19} & 1 \end{pmatrix} \quad \text{and} \quad U = \begin{pmatrix} 2 & 1 & -1 \\ 0 & \frac{19}{2} & \frac{5}{2} \\ 0 & 0 & \frac{54}{19} \end{pmatrix}$$

$$AX = I \quad \therefore LUX = I$$

$$\begin{pmatrix} 1 & 0 & 0 \\ \frac{1}{2} & 1 & 0 \\ -\frac{1}{2} & \frac{5}{19} & 1 \end{pmatrix} \cdot \begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \Rightarrow \begin{matrix} y_1 = 1; \frac{1}{2}y_1 + y_2 = 0 \\ y_2 = -\frac{1}{2}; y_3 = \frac{12}{19} \end{matrix}$$

$$\vec{y} = \begin{pmatrix} 1 \\ -\frac{1}{2} \\ \frac{12}{19} \end{pmatrix}$$

$$\begin{pmatrix} 2 & 1 & -1 \\ 0 & \frac{19}{2} & \frac{5}{2} \\ 0 & 0 & \frac{54}{19} \end{pmatrix} \cdot \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 1 \\ -\frac{1}{2} \\ \frac{12}{19} \end{pmatrix} \quad \vec{x} = \begin{pmatrix} \frac{2}{3} \\ -\frac{1}{9} \\ \frac{2}{9} \end{pmatrix}$$

$$\frac{54}{19} x_3 = \frac{12}{19} \quad \therefore x_3 = \frac{2}{9} \quad \frac{19}{2} x_2 + \frac{5}{2} x_3 = -\frac{1}{2}$$

$$x_2 = -\frac{1}{9}$$

$$2x_1 - \frac{1}{9} - \frac{2}{9} = 1 \quad x_1 = \frac{2}{3}$$

$$\frac{19}{2} x_2 = \left(-\frac{1}{2} - \frac{5}{9}\right) \cdot \frac{2}{19}$$

$$\vec{x} =$$

1.2.1

$$\begin{pmatrix} 1 & 0 & 0 \\ \frac{1}{2} & 1 & 0 \\ -\frac{1}{2} & \frac{5}{19} & 1 \end{pmatrix} \cdot \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$$

$$y_1 = 0; y_2 = 1; y_3 = -\frac{5}{19}$$

$$\vec{y}_2 = \begin{pmatrix} 0 \\ 1 \\ -\frac{5}{19} \end{pmatrix}$$

$$\begin{pmatrix} 2 & \frac{1}{19} & -1 \\ 0 & \frac{19}{2} & \frac{5}{2} \\ 0 & 0 & \frac{54}{19} \end{pmatrix} \cdot \begin{pmatrix} x_1 \\ x_2 \\ -x_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \\ -\frac{5}{19} \end{pmatrix}$$

$$x_3 = -\frac{5}{54} \quad \text{L''}$$

$$\frac{19}{2}x_2 - \frac{25}{108} = 1$$

$$x_2 = \frac{2}{54}$$

$$x_1 = -\frac{1}{9}$$

$$\vec{x}_2 = \begin{pmatrix} -\frac{1}{9} \\ \frac{2}{54} \\ -\frac{5}{54} \end{pmatrix}$$

1.2.2

$$\begin{pmatrix} 1 & 0 & 0 \\ \frac{1}{2} & 1 & 0 \\ -\frac{1}{2} & \frac{5}{19} & 1 \end{pmatrix} \cdot \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix}$$

$$y_1 = 0, y_2 = 0, y_3 = 1$$

$$U \cdot \vec{y}_3 = \vec{x}_3$$

$$\begin{pmatrix} 2 & \frac{1}{19} & -1 \\ 0 & \frac{19}{2} & \frac{5}{2} \\ 0 & 0 & \frac{54}{19} \end{pmatrix} \cdot \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

$$x_3 = \frac{19}{54} \quad x_2 = -\frac{5}{54}$$

$$\vec{x}_3 = \begin{pmatrix} \frac{2}{54} \\ -\frac{5}{54} \\ \frac{19}{54} \end{pmatrix}$$

$$2x_1 + \left(-\frac{5}{54}\right) - \frac{19}{54} = 0$$

$$2x_1 = -\frac{24}{54} \quad x_1 = -\frac{2}{9}$$

$$A^{-1} = \begin{pmatrix} \frac{2}{3} & -\frac{1}{9} & \frac{2}{9} \\ -\frac{1}{9} & \frac{2}{54} & -\frac{5}{54} \\ \frac{2}{9} & -\frac{5}{54} & \frac{19}{54} \end{pmatrix}$$

12
a)

$$\begin{aligned} 5x_1 + 2x_2 + x_3 &= 7 \\ -x_1 + 4x_2 + 2x_3 &= 3 \\ 2x_1 - 3x_2 + 6x_3 &= -1 \end{aligned}$$

$$x^0 = \begin{pmatrix} -1,4 \\ 3,9 \\ 0,2 \end{pmatrix} \quad \varepsilon = 10^{-1}$$

~~ii)~~ i) $5 \neq 0; 4 \neq 0; 10 \neq 0$

ii) $\text{Det}(A_1) \neq 0$ e iii $|S| > |2| + |1|; |4| > |-1| + |2|; |10| > |-3| + |2|$
 $\therefore$ converge

$k=0$

$$x_1^{(1)} = \frac{7 - 2 \cdot 3,9 - 0,2}{5} = -0,2$$

$$x_2^{(1)} = \frac{3 + (-1,4) - 2 \cdot (0,2)}{4} = 0,3$$

$$x_3^{(1)} = \frac{-1 - 2 \cdot (-1,4) + 3 \cdot 3,9}{10} = 1,35$$

Critères de passage

$$\|x'\|_{\infty} = \max \{|-0,2|, |0,3|, |1,35|\} \Rightarrow \underline{\underline{1,35}}$$

$$\|x' - x^0\|_{\infty} = \max \{| \Delta x_1 |, | \Delta x_2 |, | \Delta x_3 | \} = 3,6$$

$CP > 0,1$

$k=1$

$$x_1^2 = \frac{7 - 2 \cdot (0,3) - 1,35}{5} = 1,01$$

$$x_2^2 = \frac{3 + (-0,2) - 2 \cdot (1,35)}{4} = 0,025$$

$$x_3^2 = \frac{-1 - 2 \cdot (-0,2) + 3 \cdot (0,3)}{10} = 0,03$$

$$P > 0,1$$

$$k=3$$

$$x_1^3 = \frac{7 - 2(0,025) - 0,03}{5} = 1,384$$

$$x_2^3 = \frac{3 + 1,01 - 2 \cdot (0,03)}{4} = 0,9875$$

$$x_3^3 = \frac{-1 - 2 \cdot (1,01) + 3 \cdot (0,025)}{10} = -0,2945$$

$$CP > 0,1$$

$$k=3$$

$$x_1^4 = \frac{7 - 2 \cdot (0,9875) + 0,2945}{5} = 1,0639$$

$$x_2^4 = \frac{3 + 1,384 - 2 \cdot (-0,2945)}{4} = 1,24325$$

$$x_3^4 = \frac{-1 - 2 \cdot (1,384) + 3 \cdot (0,9875)}{10} = -0,08055$$

$$CP > 0,1$$

$$k=4$$

$$x_1^5 = \frac{7 - 2(1,24325) + 0,08055}{5} = 0,9188$$

$$x_2^5 = \frac{3 + 1,0639 + 2 \cdot (0,08055)}{4} = 1,0563$$

$$x_3^5 = \frac{-1 - 2(1,0639) + 3 \cdot (1,24325)}{10} = 0,0602$$

$$k=5$$

$$x_1^6 = \frac{7 - 2 \cdot (1,0563) - 0,0602}{5} = 0,9655$$

$$x_2^6 = 0,0331$$

$$x_3^6 = \frac{3 + 0,9188 - 2 \cdot (0,0602)}{4} = 0,9496$$

$$k=6 \quad x_1^7 = 1,0135 ; \quad x_2^7 = 0,9748 ; \quad x_3^7 = -0,0082$$

$$\|x^7\|_{\infty} = 1,0135$$

$$\|\Delta x^k\|_{\infty} = 0,048$$

$$CP = \frac{0,048}{1,0135} \approx 0,1$$

$$\therefore X \approx \begin{bmatrix} 1,0135 \\ 0,9748 \\ -0,0082 \end{bmatrix}$$

b)

$$10x_1 + x_2 + x_3 = 10$$

$$2x_1 + 6x_2 + 8x_3 = 20$$

$$7x_1 + x_2 + 6x_3 = 20$$

$$x^0 = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \quad \epsilon = 10^{-2}$$

i) $10 \neq 0$, ii) $\text{Det}(A) \neq 0$

iii)

Columns $|10| > |2| + |7|$, $|10| > |6| + |8|$, $|10| > |2| + |1|$

iv)

$$x_1 = \frac{10 - x_2 - x_3}{10}$$

$$x_2 = \frac{20 - 2x_1 - 8x_3}{10}$$

$$x_3 = \frac{20 - 7x_1 - x_2}{10}$$

k=0

$$x_1^1 = 0,8 ; \quad x_2^1 = 1 ; \quad x_3^1 = 1,2$$

$$k=1 \quad x_1^2 = 0,778 ; \quad x_2^2 = 0,772 ; \quad x_3^2 = 1,366$$

$$k=2 \quad x_1^3 = 0,7862 ; \quad x_2^3 = 0,7516 ; \quad x_3^3 = 1,3782$$

$$k=3 \quad x_1^4 = 0,78702 ; \quad x_2^4 = 0,7402 ; \quad x_3^4 = 1,3745$$

$$K = 4$$

$$X_1^S = 0,78853; \quad X_2^S = 0,742936; \quad X_3^S = 1,37507$$

$$\|X^S\|_6 = 1,37507$$

$$\|A^K\|_\infty = 0,002796$$

$$CP < 0,01$$

$$X \approx \begin{bmatrix} 0,78853 \\ 0,742936 \\ 1,37507 \end{bmatrix} //$$

12) a)

$$4x_1 + x_2 + x_3 + x_4 = 7$$

$$2x_1 - 3x_2 + x_3 - x_4 = -6$$

$$x_1 + 2x_2 - 5x_3 + x_4 = -1$$

$$x_1 + x_2 + x_3 - 4x_4 = -1$$

$$X^0 = \begin{pmatrix} 0 \\ 0,2 \\ -0,4 \\ 0 \end{pmatrix}, \epsilon = 10^{-2}$$

i) Diagonal $\neq 0$; ii) $|4| > |1| + |1| + |1| + |1|$; $|2| > |1| + |2| + |1| + |1|$; $|-5| > |1| + |1| + |1| + |1|$; $|-4| > |1| + |1| + |1| + |1|$

Converge

~~12) a)~~

~~12) a)~~

$$x_1 = \frac{7 - x_2 - x_3 - x_4}{4}$$

$$x_2 = \frac{-6 + x_3 - x_4 + 2x_1}{8}$$

$$x_3 = \frac{-1 + x_1 + 2x_2 + x_4}{5}$$

$$x_4 = \frac{-1 + x_1 + x_2 + x_3}{4}$$

$k=0$

$$x_1^1 = 1,8 ; x_2^1 = 1,15 ; x_3^1 = 1,02 ; x_4^1 = 1,2425$$

$k=1$

$$x_1^2 = 0,896875 ; x_2^2 = 0,94640625 ; x_3^2 = 1,0064375$$

$$x_4^2 = 0,9624296875$$

CP

$$\|x^2\|_{\infty} = 1,0064375 ; \|Ax^2\|_{\infty} = \{ |0,903125|, |0,07359375|, |0,0135625|, |0,280071| \}$$

$$\frac{0,903125}{1,0064375} = 0,8973483202$$

$k=2$

$$x_1^3 = 1,02118 ; x_2^3 = 1,01078 ; x_3^3 = 1,039841938$$

$$x_4^3 = 1,001970$$

$k=3$

$$x_1^4 = 0,9950 ; x_2^4 = 0,9978 ; x_3^4 = 0,9998 ; x_4^4 = 0,9982$$

$k=4$

$$x_1^4 = 1,0011 ; x_2^4 = 1,0005 ; x_3^4 = 1 ; x_4^4 = 1,0004$$

$$X \approx \begin{pmatrix} 1,0011 \\ 1,0005 \\ 1 \\ 1,0004 \end{pmatrix}$$

$$Err = \frac{\|X^4 - X^3\|_{\infty}}{\|X^4\|_{\infty}} = \frac{10,0061}{1,0011} = 0,0060$$

$$\{0,0061, 0,0027, 0,0021, 0,0022\}$$

⑥

$$4x_1 + x_2 + x_3 = 6$$

$$x_1 + 6x_2 + x_3 = 8$$

$$2x_1 + x_2 + 8x_3 = 11$$

$$x^0 = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}, \epsilon = 10^{-2}$$

i) Diagonal dominant, i.e.) $|4| > |1| + |1|$; $|6| > |1| + |1|$; $|8| > |1| + |1|$

iii) $\det(A) \neq 0$

$$x_1 = \frac{6 - x_2 - x_3}{4}; \quad x_2 = \frac{8 - x_1 - x_3}{6}$$

$$x_3 = \frac{11 - 2x_1 - x_2}{8}$$

$K=0$

$$x_1^1 = 1,475; \quad x_2^1 = 1,0875; \quad x_3^1 = 0,8703125$$

$K=1$

$$x_1^2 = 1,010546875; \quad x_2^2 = 1,019856771; \quad x_3^2 = 0,994881849$$

$$K=2 \quad x_1^3 = 0,996315511; \quad x_2^3 = 1,001467; \quad x_3^3 = 1,000737742$$

$$K=3 \quad x_1^4 = 0,9994; \quad x_2^4 = 1; \quad x_3^4 = 1,0001$$

$$CP = \frac{0,00308}{1,0001} = 0,00307$$

$$X \approx \begin{pmatrix} 0,9994 \\ 1 \\ 1,0001 \end{pmatrix}$$